

AJA STARKEY

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PROFESSIONAL SUMMARY

Data scientist with an MS in Data Science from the University of Colorado Boulder (May 2026) and a decade of experience spanning pharmaceutical sales, clinical operations, and laboratory analytics. Combines deep healthcare and biotech domain expertise with rigorous machine learning training to build models that drive decisions in complex, high-stakes environments. Skilled at translating technical findings to diverse stakeholders and leading cross-functional initiatives from scoping through delivery.

CORE COMPETENCIES

- Machine Learning & Predictive Modeling
- NLP & Deep Learning (CNNs, Transformers, LLMs)
- Statistical Analysis & Experimental Design
- Data Visualization & Communication
- Python · R · SQL
- Clinical & Biotech Domain Expertise
- Cross-functional Stakeholder Engagement
- Project & Program Management
- Regulatory & Quality Assurance (FDA, ISO)

EDUCATION

Master of Science, Data Science

University of Colorado Boulder | May 2026

- Advanced coursework: machine learning, deep learning, NLP, statistical modeling, data mining, and predictive analytics
- Capstone projects in healthcare ML: TNBC molecular subtype classification (TCGA-BRCA), histopathologic cancer detection, medical image CNNs

Bachelor of Science, Integrated Sciences — Minor: Psychology

University of Denver | 2019

TECHNICAL SKILLS

Languages & Libraries: Python (pandas, NumPy, scikit-learn, TensorFlow, PyTorch), R (Tidyverse, ggplot2), SQL

Machine Learning: Random Forest, XGBoost, SVM, AutoGluon, CNNs, GANs, DistilBERT, transfer learning, PCA, NMF, clustering

NLP & LLMs: HuggingFace Transformers, ktrain, text classification, fine-tuning, RAG

Visualization & Reporting: Plotly, Matplotlib, Seaborn, R Markdown, data dashboards

Platforms: Google Colab, Jupyter, GitHub, VS Code

SELECTED PROJECTS

Triple-Negative Breast Cancer Molecular Subtype Classification

MS Data Science Capstone | 2025

- Classified TNBC subtypes (BL1, BL2, M, LAR) from TCGA-BRCA RNA-seq data (175 samples) using AutoGluon, Random Forest, XGBoost, and SVM
- Prioritized macro F1 over accuracy given clinical class imbalance; best result: 40% accuracy, macro F1 of 0.361
- Additional multi-omic and clinical data will improve classification; 40% accuracy exceeds random chance when biopsy RNA testing has not yet been performed

Medical Imaging CNNs — Histopathology & Skin Cancer Detection

MSDS-5511 Deep Learning | 2024–2025

- Built CNNs for binary cancer detection (Kaggle PCam) and multi-class skin lesion classification (HAM10000) using transfer learning

NLP Text Classification & LLM Fine-tuning

MSDS-5511 / DTSA-5798 | 2024–2025

- Fine-tuned DistilBERT for disaster tweet classification; achieved ~81% accuracy on health article classification using ktrain

COVID-19 & NYPD Shooting Geospatial Analysis

Data Science Foundations | 2023–2024

- Statistical and geospatial analysis of public health datasets using R, ggplot2, and log fold-change mapping
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PROFESSIONAL EXPERIENCE

Laboratory Analyst

Mesa Labs | Bozeman, MT | 2023–2024

- Ensured FDA and ISO quality assurance compliance for biological indicator testing; performed regression analysis to determine microbial inactivation kinetics
- Generated analytical dashboards and technical documentation supporting process optimization and regulatory validation

Pharmaceutical Representative

Inizio Healthcare | Bozeman, MT | 2022–2023

- Managed relationships with 300+ healthcare professionals; communicated clinical trial efficacy data and statistical outcomes to physicians and administrators
- Applied quantitative engagement metrics and data-driven strategies to optimize outreach effectiveness

Co-Director & Head Coach

Montana Endurance Academy | Bozeman, MT | 2021–2022

- Led operations and strategic planning for youth endurance academy (50+ athletes); integrated performance data into individualized training models

Program Director & Head Coach

Big Sky Ski Foundation | Bozeman, MT | 2020–2021

- Drove 5x program growth through strategic planning and evidence-based training; managed multi-site operations and cross-functional stakeholder coordination

Medical Assistant

Skin Care MT | Bozeman, MT | 2019–2020

- Supported dermatological procedures; maintained patient records, coordinated care workflows, and volunteered in Emergency Department at Swedish Hospital, Denver (2017–2019)

Research Intern

National Science Foundation REU Program | Bozeman, MT | Summer 2016

- Designed and executed stem cell research experiments; applied regression analysis to evaluate and present experimental outcomes
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NOTABLE ACHIEVEMENTS

- 2x NCAA Team National Champion, Nordic Skiing (2016 & 2018) · NCAA Division I All-American (2016) · Rossignol Sponsored Athlete (2012–2023)
- 2025 Boston Marathon finisher, 3:21 PR — reflects discipline, goal-setting, and sustained performance under pressure